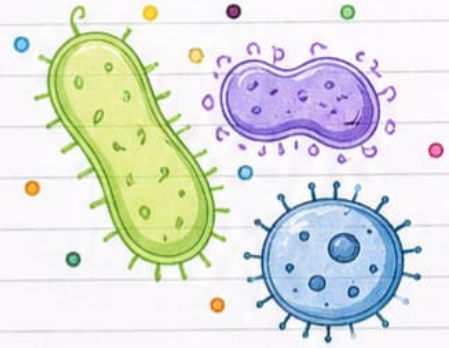


KEY CONCEPT

Microorganisms help in food production, medicine, industry & environment.

1 INTRODUCTION

- Microbes have been our silent helpers since ancient times.
- They are used in food, medicine, industry, agriculture and environmental management.
- Their ability to carry out biochemical reactions makes them very **useful**.



2 MICROBES IN FOOD PRODUCTION

Fermented Foods

Microbes convert complex compounds into simpler forms, improving **taste**, **texture** and **nutritional value**.

Examples:

- **Curd** - *Lactobacillus* converts lactose into lactic acid.
- **Idli/Dosa batter** - Fermentation by *Lactobacillus* and **yeast**.
- **Bread** - **Yeast** (*Saccharomyces*) produces CO_2 which makes dough fluffy.
- **Vinegar** - *Acetobacter* converts alcohol into **acetic acid**.
- **Cheese** - Made using bacteria, enzymes and rennet.

BENEFITS

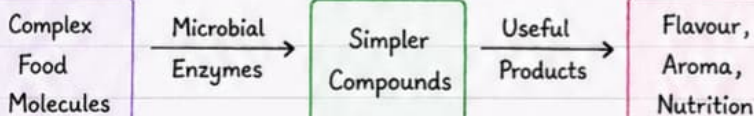
- ♥ Improved flavour
- ♥ Easy digestion
- ♥ Increased shelf life
- ♥ Enhanced nutrition

DID YOU KNOW?

Yogurt was discovered by accident when milk was stored in warm conditions!



HOW FERMENTATION WORKS ?



EXAMPLE: CURD PRODUCTION



Boil milk and cool it slightly.



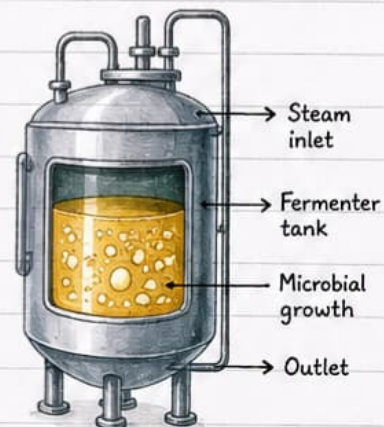
Add a spoon of curd (starter culture).



Keep in warm place for 6-8 hours.



Lactobacillus converts lactose into lactic acid → Curd formed



Industrial Fermenter

3 MICROBES IN INDUSTRY

- Microbes are used in large scale industrial processes to produce **antibiotics**, **enzymes**, **organic acids**, **biofuels** and other chemicals.



Note

Fermentation is the key process used by microbes in industries.



ANTIBIOTICS

- Penicillin - obtained from *Penicillium notatum*.
- Streptomycin - obtained from *Streptomyces griseus*.
- These antibiotics kill or inhibit harmful bacteria.



ORGANIC ACIDS

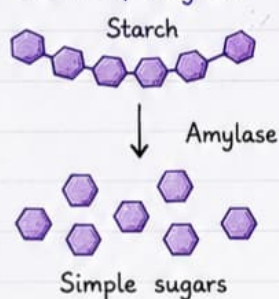
- Citric acid - *Aspergillus niger*
- Lactic acid - *Lactobacillus*
- Acetic acid - *Acetobacter*

Used in food, cosmetics, pharmaceuticals and textiles.

ENZYMES

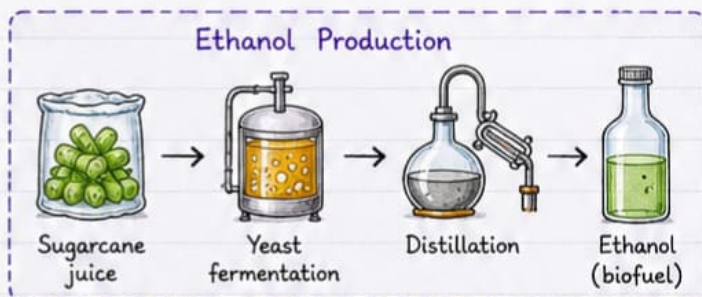
- Microbial enzymes are used in **detergents**, **food** processing, **leather** and **paper** industries.
- Example - Amylase (from *Bacillus*) breaks starch into sugars.

Action of Amylase



BIOFUELS

- Ethanol (bioethanol) is produced by yeast (*Saccharomyces cerevisiae*) through fermentation of sugars.
- Used as a renewable fuel and in alcoholic beverages.



4 MICROBES IN AGRICULTURE

BIOFERTILIZERS

- Microbes improve soil fertility by fixing nitrogen or solubilizing minerals.

Microbe	Function
Rhizobium	Fixes nitrogen in root nodules of leguminous plants.
Azotobacter	Free living nitrogen fixer.
Anabaena	Blue-green algae; fixes nitrogen in paddy fields.
Phosphate Solubilizing Bacteria (PSB)	Solubilizes bound phosphates in soil.

BIOCONTROL AGENTS

- Microbes are used to control plant pathogens (harmful fungi, bacteria, insects).
- They are eco-friendly alternatives to chemicals.

Examples:

- Trichoderma* - controls soil-borne fungi.
- Bacillus thuringiensis* (Bt) - controls insect pests.
- Nucleopolyhedrovirus (NPV) - used against specific insects.



★ IN SHORT ★

- ✓ Microbes are tiny but mighty!
- ✓ They help in producing food, medicines, fuels, chemicals and improve soil health.
- ✓ They are essential for sustainable development and a better future.

“Take care of microbes,
they take care of life!”



* Small microbes,
Big difference! *

5 MICROBES IN MEDICINE



- Microbes are the backbone of many **life-saving** medicines and treatments.

ANTIBIOTICS

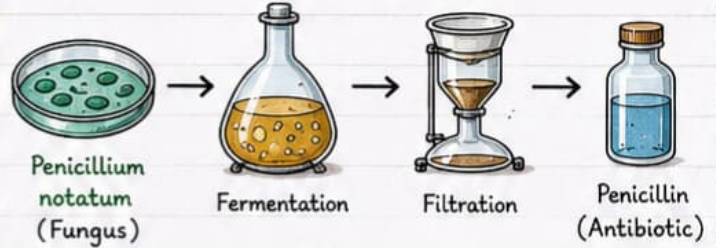
- Penicillin - from *Penicillium notatum*
- Streptomycin - from *Streptomyces griseus*
- Chloramphenicol, Tetracycline, Erythromycin - from different microbes.
- These drugs kill or stop the growth of harmful bacteria.

Did You Know?

The antibiotic **Penicillin** was discovered by **Alexander Fleming** in **1928** from *Penicillium notatum*.



Production of Penicillin



VACCINES

- Vaccines contain weakened or killed microbes or their products.
- They help our body develop immunity against diseases.
- Example: **BCG vaccine** (from *Mycobacterium bovis*) protects against tuberculosis.

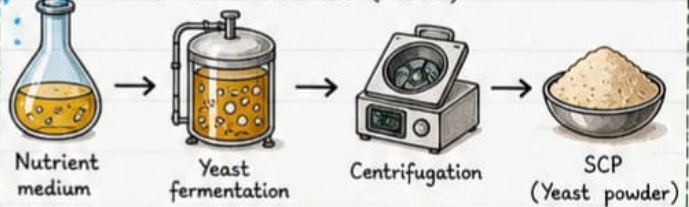


Vaccines train the immune system to fight infections!

SINGLE CELL PROTEIN (SCP)

- Microbes like **yeast**, **bacteria**, and **algae** are rich in protein.
- They are used as food supplements for humans and animal feed.
- Example: Yeast (*Saccharomyces cerevisiae*) is used as SCP.

SCP Production (Yeast)



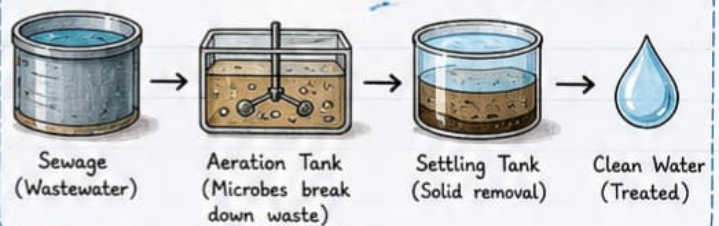
6 MICROBES IN ENVIRONMENTAL MANAGEMENT



BIODEGRADATION OF WASTES

- Microbes break down complex organic wastes into simpler substances.
- They are used in **sewage treatment**, **composting**, and **recycling** of wastes.
- Example: **Bacteria** and **fungi** degrade garbage, sewage, and industrial wastes.

Sewage Treatment by Microbes



BIOREMEDIATION

- Microbes are used to clean up pollutants from soil, water and air.
- They degrade harmful chemicals, oils, pesticides and other toxic compounds.
- Example: Oil-eating bacteria (*Pseudomonas*) help clean oil spills.



KEY TAKEAWAY

- ✓ Microbes are nature's tiny workers.
- ✓ They help in producing food, medicines, industrial products and clean our environment.
- ✓ Without microbes, life as we know it would not be possible!



“Small microbes, Big impact!”



7 TRADITIONAL MICROBES & MODERN BIOTECHNOLOGY

- Microbes have been used traditionally for centuries in making various **foods** and **beverages**.

REMEMBER

Humanity has used microbes long before we knew about them. This knowledge is called **traditional knowledge**.



A. TRADITIONAL FOODS

Food	Microorganism	Use / Role
Curd (Yogurt)	<i>Lactobacillus</i>	Converts milk lactose into lactic acid; preserves milk.
Idli/Dosa batter	<i>Lactobacillus</i> , yeast	Fermentation makes batter soft and fluffy.
Bread	Yeast (<i>Saccharomyces</i>)	Produces CO ₂ gas that makes dough rise.
Cheese	<i>Lactobacillus</i> , fungi	Help in curdling milk and develop flavour.
Vinegar	<i>Acetobacter</i>	Converts alcohol into acetic acid.
Soy sauce	<i>Aspergillus</i> , yeast	Fermentation of soybeans.
Tempeh	<i>Rhizopus</i>	Fermentation of soybeans; rich in protein.

B. TRADITIONAL BEVERAGES

Beverage	Microorganism	Use / Role
Alcoholic beverages (Beer, Wine)	Yeast (<i>Saccharomyces</i>)	Yeast ferments sugars and produces alcohol and CO ₂ .
Toddy (Palm wine)	Yeast	Natural fermentation of sap.
Kefir	Bacteria + Yeast	Fermentation of milk; probiotic drink.
Kombucha	Bacteria + Yeast	Fermented tea; rich in probiotics.

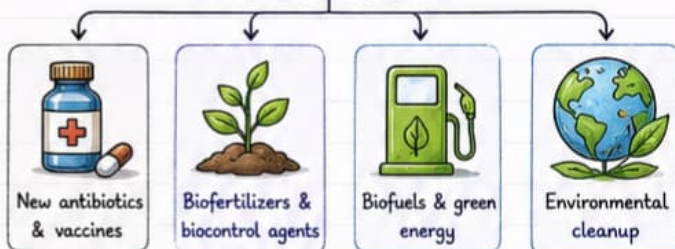
★ Fun Fact

Curd, cheese, bread, wine – all are ancient biotechnology made using microbes!

8 FUTURE PROSPECTS

- Microbes are a **treasure trove** of potential.
- They can help solve global problems like **diseases**, **pollution** and **food security**.
- Ongoing research in microbiology and biotechnology will lead to new solutions in the future.

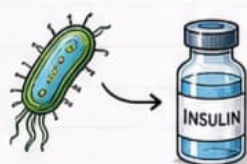
Areas of Future Use



EXAMPLES OF MICROBIAL APPLICATIONS

1. Production of Insulin

Bacteria (*E. coli*) are used to produce human insulin for diabetics.



Genetically engineered bacteria

2. Bioleaching

Bacteria (*Thiobacillus ferrooxidans*) extract metals from ores.



Metal

3. Bioplastics

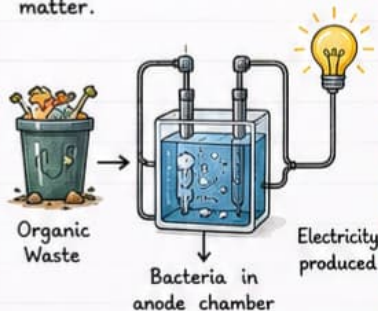
Bacteria produce biodegradable plastics (e.g., PHB).



Eco-friendly plastic

4. Microbial Fuel Cells

Bacteria generate electricity while breaking down organic matter.



Electricity produced

QUICK REVISION

- ✓ Microbes help in **food**, **medicine**, **industry**, **agriculture** & **environment**.
- ✓ They are used to produce curd, bread, antibiotics, enzymes, biofuels.
- ✓ They improve soil fertility, control pests and clean pollution.
- ✓ Traditional knowledge + modern science = a better future.
- ✓ Microbes are **small**, but their impact is **huge**!

THINK & SMILE

"The world is run by microbes. We just live in it."

– Unknown



Keep Learning, Keep Exploring!

MICROBES – NATURE'S INVISIBLE HEROES